

Demonstration of MorphKeys: A Reconfigurable Keyboard System Using Switched NFC Tags

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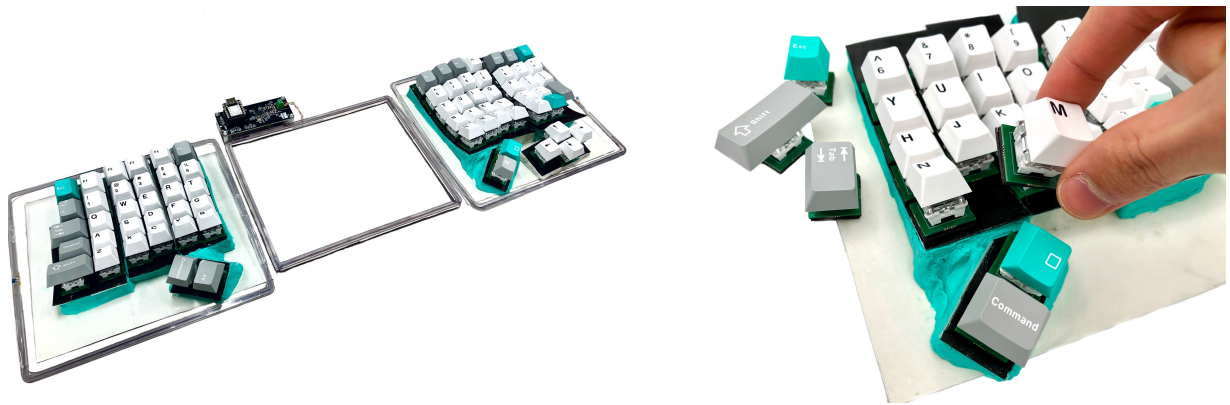


Figure 1: Overview of MorphKeys. MorphKeys is a reconfigurable keyboard system that allows key arrangements to be changed based on the user and use case. Each key is a battery-free unit powered and scanned wirelessly.

Abstract

We demonstrate MorphKeys, a reconfigurable keyboard system that lets users freely arrange individual keys in three dimensions to match their ergonomic needs and specific usage scenarios. Each key is an independent, battery-free module powered and read via near-field communication (NFC) only while being pressed, enabling wireless key input detection. The system achieves an extended reading range by incorporating relay resonators in the NFC reader. Keys are magnetically fixed to a custom base made of clay and iron sand, allowing flexible, stable placement without interfering with NFC operation. In this demonstration, we show how MorphKeys enables personalized key layouts while maintaining full keyboard functionality.

CCS Concepts

• **Human-centered computing** → **Keyboards.**

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1 Introduction

Keyboards are widely used as an input device for computers, but their shapes and key layouts are often not optimized for human use [2]. Even keyboards designed based on ergonomics cannot fully accommodate individual differences such as hand size, personal preferences, or varying use cases.

Although there has been research on keyboards that allow users to physically change the key layout, prior work has not achieved both a sufficient number of keys (approximately 60–100) and a three-dimensional key arrangement, due to limitations in power consumption or communication range [1, 3–5].

We present MorphKeys [6], a reconfigurable keyboard system that enables users to freely and three-dimensionally customize the physical key layout (Figure 1). MorphKeys employs battery-free

key units based on switched NFC tags and an NFC reader system utilizing relay resonators. Each key is magnetically attached to a three-dimensional base made of clay and iron sand. When the user presses a key, it functions as an NFC tag, and the reader converts its unique ID into a key input. This design allows users to freely arrange keys in three dimensions because each unit is wirelessly powered and communicates without wiring restrictions.

2 Design of MorphKeys

MorphKeys consists of three main components: 1) battery-free key units that function as NFC tags only while pressed by the user, 2) a three-dimensional base that uses magnets to hold the key units in place while allowing the user to change the shape freely, and 3) a reader that powers and reads the key units.

The working principle is as follows. First, the user presses a key. This action electrically connects the tag IC and the coil, enabling the key unit to function as an NFC tag. Next, the reader detects the tag according to the NFC protocol. When the reader discovers a tag, it indicates that the corresponding key is being pressed. The reader then sends the key input associated with the tag's unique ID over USB. If multiple keys are pressed simultaneously, tag responses may collide; however, since the NFC protocol includes an anti-collision mechanism that reads multiple tag IDs sequentially, the system can properly detect multiple simultaneous key presses. Also, we incorporate relay resonators [7] in the NFC reader to achieve a sufficient reading range for a 3D keyboard while maintaining reading performance and power consumption.

3 Application

The ability to freely customize key layouts in MorphKeys enables a wide range of interactions. We present two primary application examples.

3.1 User-specific Key Layouts

A key application of MorphKeys is user-specific key layouts. With MorphKeys, users can freely modify and adjust the key arrangement based on their physical characteristics such as hand size, or their personal preferences, including frequently used shortcuts (Figure 2). MorphKeys allows for three-dimensional adjustment of all keys, including key spacing and rotation angle, and also enables changes to the overall layout. This characteristic enables the creation of an ergonomic keyboard tailored to the individual user. Unlike custom-built keyboards created with tools like 3D printers, MorphKeys allows users to arrange keys more intuitively. It is also easy to explore various layouts and iteratively refine the layout through prolonged daily use.

3.2 Application-specific Key Layouts

MorphKeys enables key layouts tailored to specific use cases. Computers are used for a wide range of tasks, such as document writing, spreadsheet editing, and CAD operation—each of which benefits from different key arrangements. For example, in CAD applications such as KiCAD, keyboards are primarily used for executing various shortcuts and numeric input, and text input is rarely required. MorphKeys enables the optimization of key arrangements for such



Figure 2: Examples of user-specific key layouts. Users can freely modify the layout based on their physical characteristics and preferences.

application-specific use cases. Figure 3 illustrates a key layout designed for CAD operations, where frequently used shortcuts and numeric keys are placed in easily accessible positions. Moreover, MorphKeys can function as a compact macro pad by operating solely with the source reader. Combined with the previously discussed capabilities—such as the three-dimensional arrangement of keys in ergonomically convenient positions, and the ability to add or reposition keys over time—this enables novel interactions that are difficult to achieve with conventional keyboards or macro pads.



Figure 3: Example of a key layout for CAD operation. Users can freely modify the layout based on their use cases and tasks.

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